

StockFlow

Digital Inventory & Ration Ordering System

A single, presentation-ready overview — written for a senior, non-technical audience.

Prepared by SJCEM · StockFlow v1.1.0 · 13 July 2026

1. The Problem We Are Solving

Naval mess and canteen stores today run largely on **manual, paper-based processes**, which creates five recurring difficulties:

#	Problem	Impact
1	No live visibility of stock	Staff and customers cannot see what is actually left — items get over-committed after they are already exhausted
2	Ration entitlement tracked by hand	Each officer/rating has a fixed daily ration scale (RIK — Ration in Kind). Tallying it manually per person, per week is slow and error-prone
3	No systematic record of stock changes	Restocks, issues and adjustments are not consistently logged, making reconciliation and audit difficult
4	Slow, informal communication	Order windows and order status are announced by word of mouth or notice boards
5	Manual reporting	Stock, consumption and order reports are compiled by hand — time-consuming and inconsistent

StockFlow replaces this entire manual loop with one connected digital system: the admin manages stock and opens a live weekly ordering window on a **website**, customers order through a **mobile app** that automatically enforces their exact ration entitlement, and every action is recorded and reportable — instantly, transparently, without paperwork.

2. The System at One Glance

StockFlow has **three connected parts** working together at all times:

- **The Admin Website** — where the store in-charge loads stock, sets ration limits, opens the weekly ordering window, manages orders and downloads reports.
- **The Customer Mobile App** — where every customer gets notified, sees live stock, orders within their allowed limit, and tracks their order.
- **The Central System** — the invisible link between the two: it keeps the stock count live, enforces the ration limits automatically, and saves every record for reports.

StockFlow — How the System Works

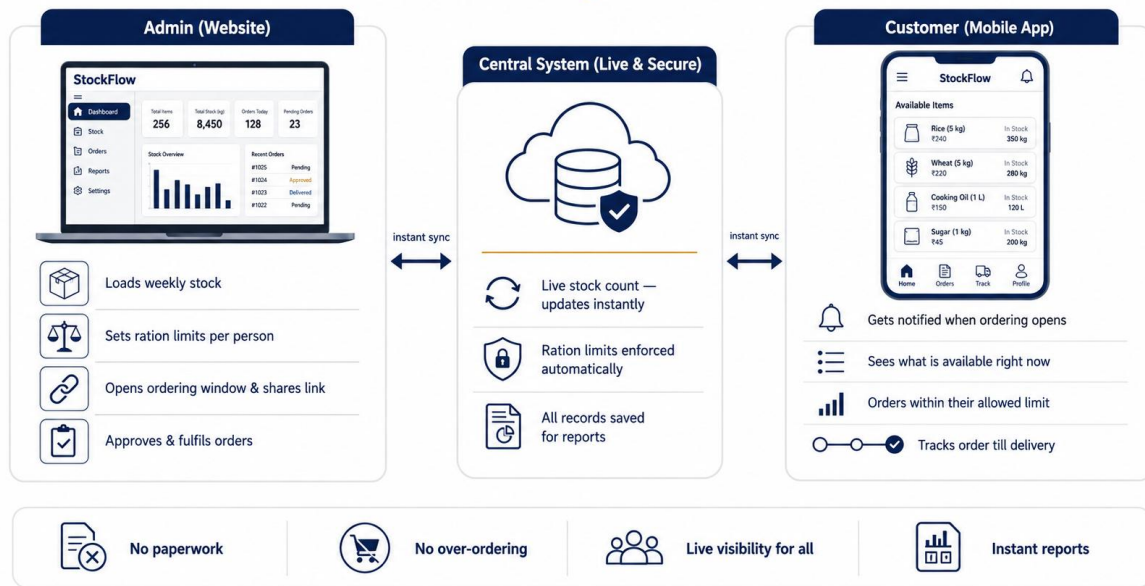


Figure 1 — The complete system at one glance.

3. Roles — Who Does What

Role	Where they work	What they do
Admin (store in-charge)	Website (full control)	Loads master stock, sets ration limits per zone, opens/closes weekly ordering links, manages users, approves & fulfils orders, sends announcements, downloads reports
Staff	Website / App (limited view)	Assists with stock updates and order handling under the admin
Customer (officer / mess / unit)	Mobile App	Receives the order notification, views live stock, places orders within their ration limit, tracks order status, views past orders

4. How the Admin Runs the Store — Website Flow

A normal week on the admin website, in six simple steps:

- Load Stock** — add items one by one, or upload the entire stock sheet (Excel/CSV) in one go.
- Set Ration Limits** — fix how much each person is allowed to order, exactly as per the official entitlement scale.
- Open the Order Window** — generate a weekly ordering link and share it with all customers.

4. **Notify Everyone** — one click sends the announcement through in-app alert, SMS, WhatsApp and email.
5. **Manage Orders Live** — watch orders arrive in real time; approve, fulfil or cancel with a tap.
6. **Download Reports** — get stock, order and shortage reports as ready PDF files.

Admin Website — Managing the Store Digitally

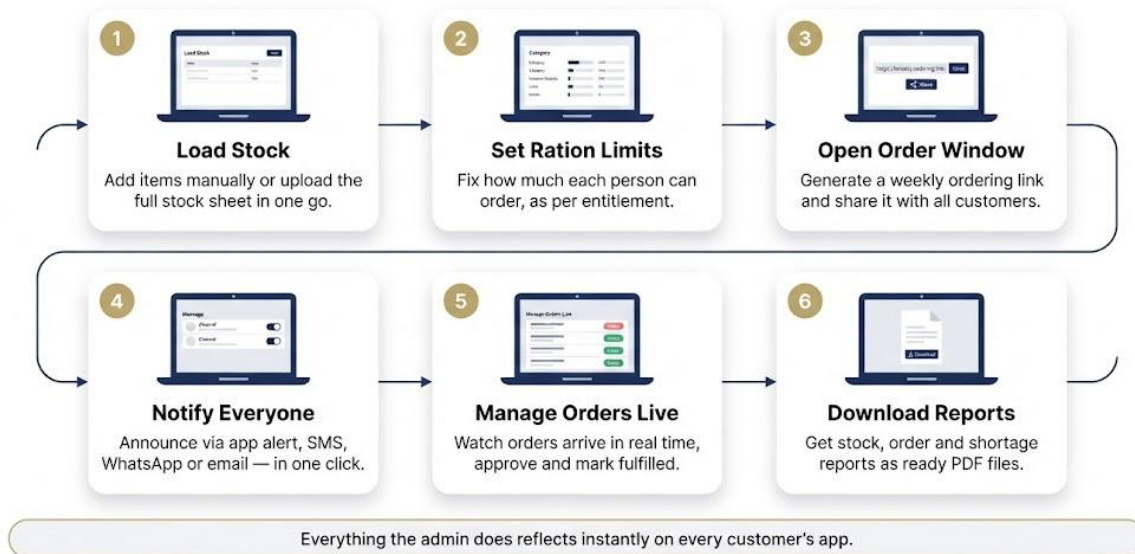


Figure 2 — Admin website flow, step by step.

5. How the Customer Orders — App Journey

The customer's experience on the mobile app, in five simple steps:

7. **Get Notified** — the moment the admin opens the window, every customer is informed.
8. **See Live Stock** — the app shows exactly what is available right now, updated live for everyone.
9. **Order Within Limit** — the app itself prevents any order beyond the customer's allowed ration; a clear limit bar shows how much of the allowance is used.
10. **Place Order** — one tap to submit. No forms, no paperwork.
11. **Track Till Fulfilled** — the order moves visibly through *Placed* → *Confirmed* → *Fulfilled*, and the full week-wise history stays saved.

Customer App — Ordering Made Simple & Fair

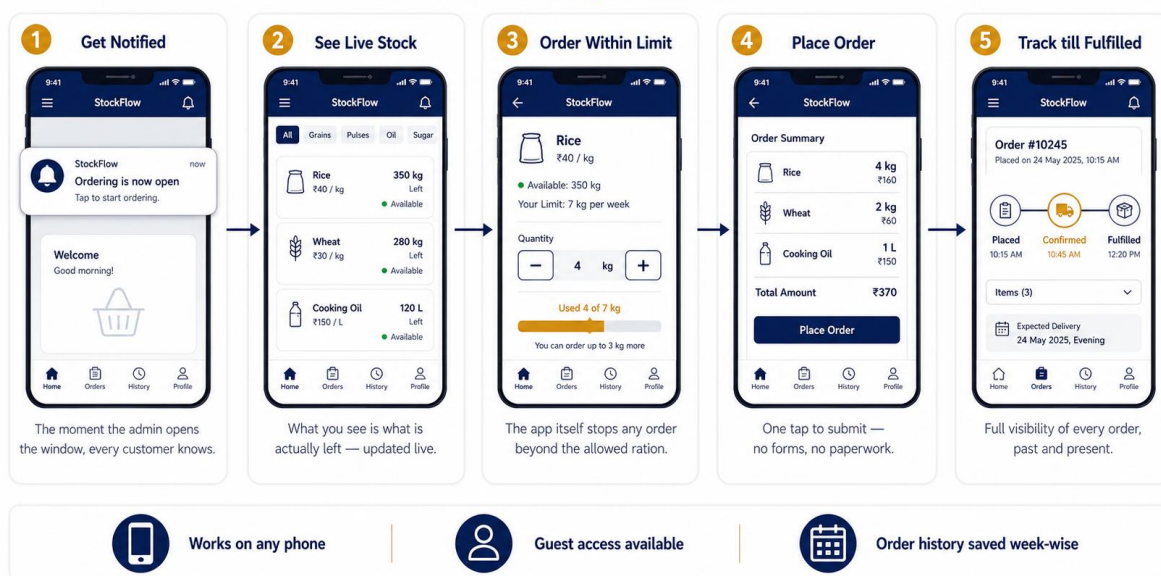


Figure 3 — Customer app journey, step by step.

6. What Is Already Built (Current Features)

6.1 Admin Website

- **Live Dashboard** — order-window status, stock alerts (low / out of stock), key numbers and recent orders on one screen, with a copyable order link.
- **Master Stock Management** — full item catalogue with search, category filters, quick restock, and reorder-level alerts.
- **Bulk Excel/CSV Import** — upload the entire stock sheet with a downloadable template, preview, and one-click apply.
- **Ration Entitlement Engine (RIK)** — the official entitlement scale (17 categories: cereals, dal, oil, sugar, milk, meat, vegetables, eggs, tea, fruit and more) is built into the system with per-person daily quantities and approved substitute ("in-lieu") articles. Admins can fine-tune overall, category-wise and item-wise limits per zone.
- **Weekly Order Links** — create multiple ordering windows, each optionally scoped to a ration zone; open, close or reopen anytime; see every order placed against each link.
- **Order Management** — all orders with status filters and one-tap transitions (pending → confirmed → fulfilled).
- **User Management** — add staff and customers; assign each customer to a ration zone.
- **PDF Reports** — stock report, low-stock/reorder report and orders report, downloadable instantly.
- **One-Click Announcements** — notify all customers through in-app alert, SMS, WhatsApp and email together, with delivery confirmation.

6.2 Customer Mobile App

- **Simple Onboarding** — guided first-run intro; register with name, unit, phone and ration scale — or continue as a guest.
- **Smart Home Screen** — order-window status, live catalogue preview, personal ordering insights (top items, weekly trends) and quick "order again" shortcuts.
- **Ration-Aware Ordering** — live stock with category filters; the master-ration bar and category quota bars make it impossible to exceed the entitlement; approved substitutes are clearly flagged.
- **Order History** — every order saved week-wise with live status.
- **Live Sync & Alerts** — stock and order status stay current automatically; notifications arrive even when the app is in the background.

7. Behind the Scenes — Simple, Secure, Reliable

The technology is kept deliberately simple to explain:

- **One central database** — a single source of truth for stock, orders and users; no duplicate registers.
- **Live sync** — any change made anywhere reaches every screen within seconds.
- **Role-based access** — admin, staff and customers each see only what they are permitted to; enforced at the data level, not just the screen level.
- **Automatic audit record** — every stock change is logged with who, what and when — ready for audit.
- **One codebase, every device** — the same system runs as a website for the admin and as an Android app for customers.

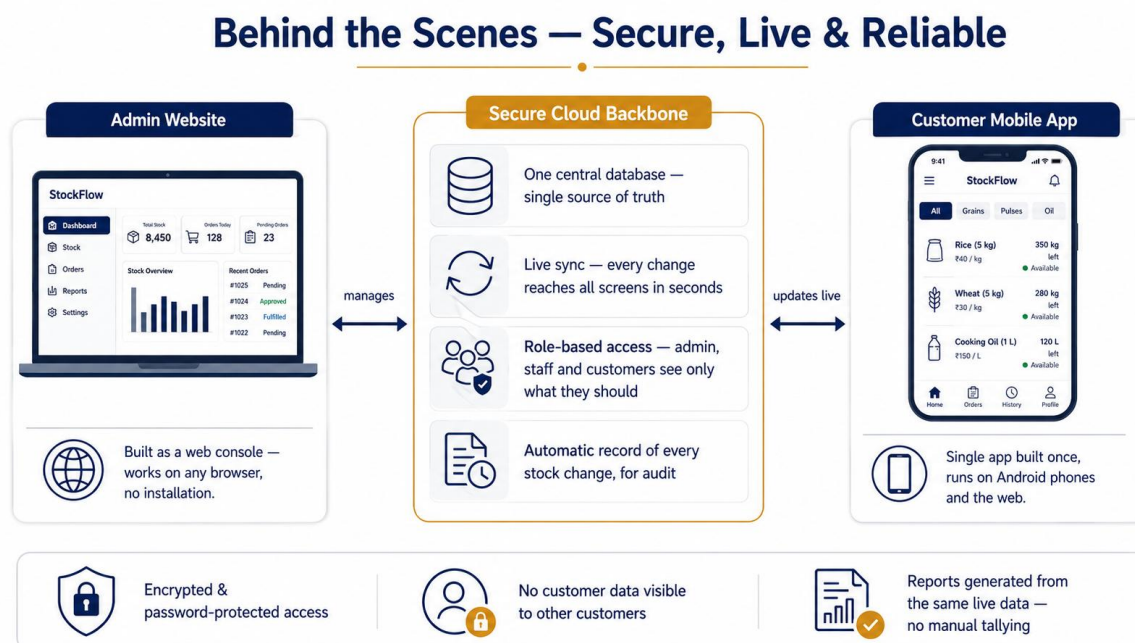


Figure 4 — Behind the scenes, kept simple.

8. The Road Ahead (Proposed Enhancements)

Priority	Enhancement	Value
High	All ration tiers/ranks — extend the entitlement engine beyond the Officers scale to every rank and mess category	Full-scale deployment readiness
High	Stock-out prediction — automatic "X will finish in Y days" alerts based on consumption rate	Prevent shortages before they happen
High	Offline mode — ordering and stock updates that work without continuous internet and sync later	Essential for ships and low-connectivity locations
Medium	Server push notifications — alerts that reach customers even when the app is fully closed	No missed order windows
Medium	OTP-based login — phone-number OTP instead of passwords for customers	Simpler and more secure onboarding
Medium	Approval workflow — supervisor sign-off for unusual or oversized orders	Extra control layer for a defence environment
Medium	Audit trail screen — full visual history of every stock change, exportable for inspections	Defence-grade accountability
Future	Multi-unit support — several messes/ships/units under one system with separate stock and reports	Scale across the organisation
Future	Advanced analytics — consumption forecasting, seasonal patterns, budget tracking	Data-driven procurement planning
Future	Hardened deployment — option to run on a restricted/local network with ID-card-based access	Meets stricter defence IT requirements

9. In One Line

StockFlow turns a paper-driven ration store into a live, fair and fully accountable digital system — stock loaded once, ordered within entitlement, tracked to the last kilogram, and reported in one click.